

Changing the Diameter in Kronecker Product of Cycles

Jeng-Jung Wang

Department of Information Engineering

I-Shou University

Kaohsiung 84008, Taiwan, Republic of China

Tung-Yang Ho*

Department of Industrial Engineering and Management

Ta Hwa Institute of Technology

Hsinchu 30740, Taiwan, Republic of China

Ting-Yi Sung

Institute of Information Science

Academia Sinica

Taipei 11529, Taiwan, Republic of China

Abstract

Communication speed in a network is related to its diameter. The diameter variability arises from the changes of links from networks. In this paper we study how the diameter of Kronecker product of cycles changes by adding edges.

1 Introduction and Definitions

An interconnection network connects the processors of a parallel and distributed system. The topology of an interconnection network for a parallel and distributed system can be represented by a loopless undirected graph, where each vertex represents a processor and each edge represents a communication link between two processors. Communication is a critical issue in the design of a parallel and distributed system. Diameter of a graph is an important factor for communication as it determines the maximum communication delay between any pair of processors in a network. To expedite communication, a smaller diameter is preferred.

In this paper, a network is represented as a loopless undirected graph. For graph definitions and notations we follow [1]. $G = (V, E)$ is a graph if V is a finite set and E is a subset of $\{(u, v) \mid (u, v) \text{ is an unordered pair of } V\}$. We say that $V(G)$ is the *vertex set* and $E(G)$ is the *edge set*. Let u and v be two different vertices in a graph G . We say that u and v are *adjacent* if $(u, v) \in E(G)$. A *path* from x_0 to x_k , denoted as $\langle x_0, x_1, x_2, \dots, x_k \rangle$, is an ordered sequence of distinct vertices such that x_i and x_{i+1} are adjacent for $0 \leq i \leq k-1$. A *cycle* is a path with at least three vertices such that the first vertex is the same as the last one. The length of a path(cycle) is the number of edges in it. The *distance* between u and v in G , denoted as $d_G(u, v)$, is the length of a shortest path joining them. The *diameter* of a graph G , denoted as $D(G)$, is the maximum distance between any two vertices. Let u and v be two different vertices in G . We say that v is a *farthest neighbor* of u if $d_G(u, v) = D(G)$.

The diameter of a graph can be affected by adding or deleting edges. Graham and Harary [3] introduced three measurements about diameter variability as follows:

*Corresponding author: hoho@thit.edu.tw.

$D^-(G)$: the least number of edges whose addition to G decreases the diameter;

$D^0(G)$: the maximum number of edges whose deletion from G does not change the diameter;

$D^+(G)$: the least number of edges whose deletion from G increases the diameter.

For example, let m -cycle C_m be a graph with vertex set $\{0, 1, 2, \dots, m-1\}$ and edge set $\{(i, i+1) | 0 \leq i \leq m-1\}$, where addition is in integer modulo m . It is known that $D(C_m) = \lfloor m/2 \rfloor$. Let P_m be a graph with vertex set $\{0, 1, 2, \dots, m-1\}$ and edge set $\{(i, i+1) | 0 \leq i \leq m-2\}$, it is known that $D(P_m) = m-1$. It is easy to check that deleting any edge renders C_m to a path of m vertices and then the diameter increases to $m-1$. On the other hand, a cycle C_m can be obtained by adding the edge $(0, m-1)$ to path P_m . The diameter decrease to $\lfloor m/2 \rfloor$. Hence, $D^-(P_m) = 1$, $D^+(P_m) = 1$, and $D^0(P_m) = 0$. The n -dimensional hypercube, Q_n , consists of all n -bit binary strings as its vertices. Two vertices are adjacent if they differ only in one bit position. Graham and Harary [3] showed that $D^-(Q_n) = 2$, $D^+(Q_n) = n-1$ and $D^0(Q_n) \geq (n-3)2^{n-1} + 2$. Bouabdallah et al. [2] improved the lower bound of $D^0(Q_n)$ and furthermore gave an upper bound $(n-2)2^{n-1} - \binom{n}{\lfloor n/2 \rfloor} + 2 \leq D^0(Q_n) \leq (n-2)2^{n-1} - \lceil (2^n - 1)/(2n - 1) \rceil + 1$.

The topology of an $m \times n$ diagonal mesh network can be represented as a Kronecker product (also known as *direct product* or *tensor product*), of two odd cycles C_m and C_n . The $m \times n$ diagonal mesh introduced by Tang and Padubidri [7] is an attractive network, where m and n are odd integers. It is a degree-4, vertex-transitive, wirable, and scalable network [5, 7]. Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be two graphs. The *Kronecker product* of G_1 and G_2 , denoted by $G_1 \otimes G_2$, is a graph in which the vertex set is defined by $V_1 \times V_2$ and the edge set is defined by $\{((u_1, u_2), (v_1, v_2)) \mid (u_1, v_1) \in E_1 \text{ and } (u_2, v_2) \in E_2\}$.

For two connected graphs G_1 and G_2 , $G_1 \otimes G_2$ is connected if and only if G_1 or G_2 contains an odd cycle [8]. Hence, in this paper, for an $m \times n$ diagonal mesh network we assume that at least one of m and n is odd and use $C_m \otimes C_n$ to represent an $m \times n$ diagonal mesh network. Without loss of generality, we assume that $m \leq n$ in this paper.

For m and n are odd, the diameter of an $m \times n$ diagonal mesh network has been studied in [7]. It is also can be derived from the formula for the distance between two different vertices in the same component of the Kronecker product of two graphs [6]. When one of m and n is odd, the diameter of $C_m \otimes C_n$ can be derived from the formula $\max\{d_{C_m}(a, b), d_{C_n}(x, y)\}$ for the shortest path between two vertices (a, x) and (b, y) that belong to the same connected component of $C_m \otimes C_n$ [4].

In this paper, we study the diameter variability arising from the changes of edges for $m \times n$ diagonal mesh network. We provide an upper bound for $D^-(C_m \otimes C_n)$. We employ the concept of *farthest neighbors* of any vertex in $C_m \otimes C_n$. In order to obtain the farthest neighbors of any vertex in the Kronecker of cycles, we compute the diameter of $C_m \otimes C_n$ using the different approach from that in [4, 6, 7]. In order to reduce the diameter of $C_m \otimes C_n$, we construct a supergraph of $C_m \otimes C_n$ by adding edges.

2 Preliminaries

Given a cycle $C_m = \langle 0, 1, \dots, m-1 \rangle$, let $N_m(i)$ denote the set of the vertices in C_m that can be reached from vertex 0 at the i th step. Then

$$N_m(i) = \begin{cases} \{1, 3, \dots, i, m-i, m-i+2, \\ \dots, m-1\} & \text{for } i \text{ odd;} \\ \{0, 2, 4, \dots, i, m-i, m-i+2, \\ \dots, m-2\} & \text{for } i \text{ even.} \end{cases}$$

For example, given C_{11} , we have $N_{11}(1) = \{1, 10\}$, $N_{11}(2) = \{0, 2, 9\}$, $N_{11}(4) = \{0, 2, 4, 7, 9\}$, $N_{11}(5) = \{1, 3, 5, 6, 8, 10\}$, $N_{11}(10) = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

With careful observation, $i = \lfloor m/2 \rfloor$ is the smallest integer satisfying $\bigcup_{j=1}^i N_m(j) = \{0, 1, 2, \dots, m-1\}$, and in particular, $N_m(\lfloor m/2 \rfloor - 1) \cup N_m(\lfloor m/2 \rfloor) = \{0, 1, 2, \dots, m-1\}$.

Since $C_m \otimes C_n$ is vertex-transitive [8], we can consider $D(C_m \otimes C_n)$ as the longest distance from $(0, 0)$ to all of the other vertices. Henceforth, we use $N(i)$ to denote the set of vertices in $C_m \otimes C_n$ that can be reached from $(0, 0)$ at the i th step. It follows that $N(i) = N_m(i) \times N_n(i)$. The diameter $D(C_m \otimes C_n)$ is the smallest integer i satisfying $\bigcup_{j=1}^i N(j) = V_m \times V_n$, where V_m and V_n are the vertex set of C_m and C_n , respectively. It can be easily verified that $D(C_m \otimes C_n) \geq \max\{\lfloor m/2 \rfloor, \lfloor n/2 \rfloor\}$.

Lemma 1 Assume that both m and n are odd. Then

$$D(C_m \otimes C_n) = \begin{cases} m-1 & \text{for } m = n, \\ m & \text{for } m+2 \leq n \leq 2m+1, \\ \lfloor n/2 \rfloor & \text{for } n \geq 2m+3. \end{cases}$$

Proof: We distinguish the following cases:

Case 1: $m = n$. Since $N_m(m-1) = V_m$ and $N_m(i) = N_n(i)$, $N(m-1) = V_m \times V_n$ and $D(C_m \otimes C_n) \leq m-1$. On the other hand, since $N_m(m-1) - N_m(m-2) = \{0\}$ and $N_m(m-2) - N_m(m-3) = \{1, m-1\}$, the vertices $(0, 1)$, $(1, 0)$, $(0, m-1)$, and $(m-1, 0)$ are not in $N(m-2)$. Therefore, $m-2 < D(C_m \otimes C_n) \leq m-1$, i.e., $D(C_m \otimes C_n) = m-1$.

Case 2: $m+2 \leq n \leq 2m+1$. Since $m \geq \lfloor n/2 \rfloor$, $N_n(m-1) \cup N_n(m) = V_n$. Furthermore, since $N_m(m) = V_m$, $N(m) = V_m \times V_n$ and $D(C_m \otimes C_n) \leq m$. On the other hand, since $N_m(m-1) - N_m(m-2) = \{0\}$, $N(m-1)$ does not contain the vertex set $\{(0, j) \mid j \in N_n(m) - N_n(m-1)\}$. It follows that $m-1 < D(C_m \otimes C_n) \leq m$, i.e., $D(C_m \otimes C_n) = m$.

Case 3: $2m+3 \leq n$, i.e., $\lfloor n/2 \rfloor \geq m+1$. Note that $N_m(\lfloor n/2 \rfloor - 1) = N_m(\lfloor n/2 \rfloor) = V_m$ and $D(C_m \otimes C_n) \geq \lfloor n/2 \rfloor$. On the other hand, since $N_n(\lfloor n/2 \rfloor - 1) \cup N_n(\lfloor n/2 \rfloor) = V_n$, $\bigcup_{i=1}^{\lfloor n/2 \rfloor} N_m(i) \times N_n(i) = V$. Thus $D(C_m \otimes C_n) = \lfloor n/2 \rfloor$. The lemma follows.

Let $F(v)$ denote the set of farthest neighbors of v in $C_m \otimes C_n$. Using the proof of Lemma 1, we can easily obtain $F(v)$ for vertex v in $C_m \otimes C_n$.

Corollary 1 Let $v = (i, j)$ be a vertex in $C_m \otimes C_n$, where both m and n are odd. Then

$$F(v) = \begin{cases} \{(i, j+1), (i, j-1), (i+1, j), (i-1, j)\} & \text{when } m = n, \\ \{(i, y) \mid y \in Y\} & \text{when } m+2 \leq n \leq 2m-1, \\ \{(i, y), (x, j+m), (x, j+m+1) \mid x \in X, y \in Y\} & \text{when } n = 2m+1, \\ \{(x, j + \lfloor n/2 \rfloor), (x, j + \lceil n/2 \rceil) \mid x \in X\} & \text{when } n \geq 2m+3, \end{cases}$$

where $X = \{0, 1, 2, \dots, m-1\}$ and $Y = \{j+1, j+3, \dots, j+(n-m-1), j+(m+1), j+(m+3), j+(m+5), \dots, j+(n-1)\}$, and both addition and subtraction are performed with modulus m in the first coordinate and with modulus n in the second coordinate.

For example, let $v = (1, 3)$ be a vertex in $C_5 \otimes C_7$, $F(v) = \{(1, 2), (1, 4)\}$. Let $v = (2, 7)$ be a vertex in $C_{11} \otimes C_{19}$, $F(v) = \{(2, 0), (2, 2), (2, 4), (2, 6), (2, 8), (2, 10), (2, 12), (2, 14)\}$. Let $v = (1, 8)$ be a vertex in $C_5 \otimes C_{13}$, $F(v) = \{(0, 1), (0, 2), (1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (3, 2), (4, 1), (4, 2)\}$.

Lemma 2 Assume that m is even and n is odd. Then $D(C_m \otimes C_n) = \begin{cases} n & \text{for } m/2 \leq n, \\ m/2 & \text{for } m/2 > n. \end{cases}$

Proof: It can be observed that each of $N_m(i)$ and $N_n(i)$ contains vertex 0 when i is even and each of $N_m(i)$ and $N_n(i)$ contains vertex 1 when i is odd.

Suppose that $m/2 \leq n$. Then,

$$N_m(n-1) = \{0, 2, 4, \dots, m-2\},$$

$$N_m(n) = \{1, 3, 5, \dots, m-1\},$$

$$N_n(n-2) = V_n - \{0\},$$

$$N_n(n-1) = N_n(n) = V_n.$$

We have $\bigcup_{i=1}^n N_m(i) \times N_n(i) = (N_m(n-1) \times N_n(n-1)) \cup (N_m(n) \times N_n(n)) = V$ and thus $D(C_m \otimes C_n) \leq n$. On the other hand, it follows from the above observation that $\bigcup_{i=1}^{n-1} N_m(i) \times$

$N_n(i) = N(n-2) \cup N(n-1) = V - N_m(m-1) \times \{0\}$, and thus $D(C_m \otimes C_n) > n-1$. Hence, $D(C_m \otimes C_n) = n$.

Suppose that $m/2 > n$. Then we have $N_n(\frac{m}{2}-1) = N_n(\frac{m}{2}) = V_n$ and $D(C_m \otimes C_n) \geq \frac{m}{2}$. On the other hand, we have $(N_m(\frac{m}{2}-1) \times N_n(\frac{m}{2}-1)) \cup (N_m(\frac{m}{2}) \times N_n(\frac{m}{2})) = V$. Hence, $D(C_m \otimes C_n) = \frac{m}{2}$. The lemma follows.

Corollary 2 *Let $v = (i, j)$ be a vertex in $C_m \otimes C_n$, where m is even and n is odd. Then*

$$F(v) = \begin{cases} \{(x, 0) \mid 1 \leq x \leq m-1 \text{ and } x \text{ is odd}\} \\ \quad \text{when } \frac{m}{2} \leq n-1, \\ \{(x, 0) \mid 1 \leq x \leq m-1 \text{ and } x \text{ is odd}\} \\ \quad \cup \{(n, y) \mid 1 \leq y \leq n-1\} \\ \quad \text{when } \frac{m}{2} = n, \\ \{(m/2, y) \mid 0 \leq y \leq n-1\} \\ \quad \text{when } \frac{m}{2} \geq n. \end{cases}$$

3 An upper bound for $D^-(C_m \otimes C_n)$

In section, we use Corollary 1 and Corollary 2 to compute $D^-(C_m \otimes C_n)$.

Theorem 1 *Assume that m and n are odd and $5 \leq m \leq n$. Then*

$$D^-(C_m \otimes C_n) \leq \begin{cases} \lfloor m/2 \rfloor \lceil n/2 \rceil \\ \quad \text{for } m \leq n \leq 2m-1, \\ 2\lfloor m/2 \rfloor \lceil n/2 \rceil \\ \quad \text{for } 2m+1 \leq n. \end{cases}$$

Proof: In order to decrease the diameter of $C_m \otimes C_n$, we add edges that connect vertex v and its farthest neighbors for v in $C_m \otimes C_n$. That is, we construct a supergraph H of $C_m \otimes C_n$ by adding a set of edges. We consider two cases as follows.

Case 1: $m \leq n \leq 2m-1$. In this case, we add the set of edges $E' = \{\langle (x, y), (x, y+1) \rangle \mid 0 \leq x \leq m-2, 0 \leq y \leq n-1 \text{ and } x, y \text{ even}\}$ to $C_m \otimes C_n$. Then, $|E'| = \lfloor m/2 \rfloor \lceil n/2 \rceil$. Let $v = (i, j)$ be an arbitrary in $C_m \otimes C_n$. Then, it suffices to show that $d_H(v, v^f) \leq D(C_m \otimes C_n) - 1$.

When $m = n$, as stated in Corollary 1, v^f can be of $(i, j+1), (i, j-1), (i-1, j)$ or $(i+1, j)$. When i, j are even, $\langle (i, j), (i, j+1), (i-1, j), (i, j+1) \rangle$ and

$\langle (i, j), (i, j+1), (i+1, j) \rangle$ are two paths of length at most 3 in H . These two paths traverses v and all of v^f . Therefore, $d_H(v, v^f) \leq 3$. Similarly, we can show that $d_H(v, v^f) \leq 3$ when at least one of i and j is odd.

When $m+2 \leq n \leq 2m-1$, v^f is stated in Corollary 1. Similar to the above case of $m = n$, we can show that $d_H(v, v^f) \leq 3$.

Case 2: $n \geq 2m+1$. Let $E'' = \{\langle (x, y), (x + \lfloor m/2 \rfloor, y + \lfloor n/2 \rfloor) \rangle, \langle (x, y+1), (x + \lfloor m/2 \rfloor, y + \lfloor n/2 \rfloor) \rangle \mid 0 \leq x < m-1, 0 \leq y \leq \lfloor n/2 \rfloor \text{ and } x, y \text{ even}\}$. Then, $|E''| = \lfloor m/2 \rfloor \lceil n/2 \rceil$. In this case, a supergraph H of $C_m \otimes C_n$ is constructed by adding $E' \cup E''$. Similar to Case 1, we can show that the diameter of H' is reduced and we omit the proof here.

Theorem 2 *Assume that m is even and n is odd. Then*

$$D^-(C_m \otimes C_n) \leq \begin{cases} \lfloor m/2 \rfloor \lceil n/2 \rceil & \text{for } m/2 < n-1, \\ \lfloor m/2 \rfloor n & \text{for } m/2 \geq n-1. \end{cases}$$

Proof: We construct a supergraph of $C_m \otimes C_n$ by adding edges between vertex $v = (i, j)$ and its farthest neighbors for $v \in C_m \otimes C_n$. And we distinguish the following cases.

Case 1. $m/2 < n-1$. As stated in Corollary 2, this case is similar to the case $m+2 \leq n < 2m+1$ in the proof of Lemma 1, hence, we add the set of edges $E' = \{\langle (x, y), (x, y+1) \rangle \mid 0 \leq x \leq m-2, 0 \leq y \leq n-1 \text{ and } x, y \text{ even}\}$ to $C_m \otimes C_n$. Then, $|E'| = \lfloor m/2 \rfloor \lceil n/2 \rceil$. It is available to decrease the diameter of $C_m \otimes C_n$.

Case 2. $m/2 = n-1$. In this case, the diameter of $C_m \otimes C_n$ is n . It can be checked that the diameter is not available to decrease when the set of edges E' is added to $C_m \otimes C_n$. The reason is that the distance between (x, y) and $(x + m/2, y)$ is still n for some even y with $0 \leq y \leq n-1$ and $0 \leq x \leq m-1$. Hence, we add another set of edges E'' to $C_m \otimes C_n$ to decrease the diameter of $C_m \otimes C_n$. Let $E'' = E' \cup \{\langle (x, y), (x + m/2, y+2) \rangle \mid 0 \leq x \leq m-2, 0 \leq y \leq n-2 \text{ and } x, y \text{ even}\}$. Then, $|E''| = \lfloor m/2 \rfloor \lceil n/2 \rceil + \lfloor m/2 \rfloor \lfloor n/2 \rfloor = \lfloor m/2 \rfloor n$ and

the distance between (i, j) and $(i + m/2, j)$ is less than or equal to 5.

Case 3. $m/2 = n$. By following the Corollary 2, the distance between (n, y_1) and (n, y_2) is less than n by adding the edges $\langle (n, y), (n, y + 1) \rangle$ to $C_m \otimes C_n$ for $0 \leq y \leq n - 1$ with y even. Hence, adding the edges in E'' is available to decrease the diameter of $C_m \otimes C_n$.

Case 4. $m/2 \geq n$. Using the similar arguments as mentioned in Case 2, adding the edges in E'' is available to decrease the diameter of $C_m \otimes C_n$.

This theorem holds.

4 Conclusion

In this paper, we have studied the diameter variability arising from the change of edges for the graph $C_m \otimes C_n$. We conjecture that the upper bounds for $D^-(C_m \otimes C_n)$ in Lemma 1 and Lemma 2, respectively, are the exact value of $D^-(C_m \otimes C_n)$.

In the future, we are interested in considering the extent of the change of diameter of graphs or graph products, i.e., given an arbitrary positive integer k , we define

$D^{-k}(G)$: the least number of edges whose addition to G decreases the diameter by at least k ;

$D^0(G)$: the maximum number of edges whose addition to G does not change the diameter;

$D^{+0}(G)$: the maximum number of edges whose deletion from G does not change the diameter;

$D^{+k}(G)$: the least number of edges whose deletion from G increases the diameter by at least k .

It is easy to check that deleting an edge from cycle C_m , increases the diameter from $\lfloor m/2 \rfloor$ to $m - 1$, we say $D^{+1}(C_m) = D^{+2}(C_m) = D^{+(m-1-\lfloor m/2 \rfloor)}(C_m) = 1$. For any graph G , $D^{+i}(G) \leq \kappa'(G)$ for all positive integer i , where

$\kappa'(G)$ is the edge connectivity of G . Note that $0 \leq D^{-i}(G) \leq D^{-j}(G)$ and $0 \leq D^{+i}(G) \leq D^{+j}(G)$ if $i \leq j$.

References

- [1] J. A. Bondy and U. S. R. Murty, *Graph theory with applications*, North Holland, New York, 1980.
- [2] A. Bouabdallah, C. Delorme, and S. Djelloul, Edge deletion preserving the diameter of the hypercube, *Discrete Applied Mathematics* 63 (1995), 91–95.
- [3] N. Graham and F. Harary, Changing and unchanging the diameter of a hypercube, *Discrete Applied Mathematics* 37/38 (1992), 265–274 .
- [4] P. K. Jha, S. Klavžar, and B. Zmazek, Isomorphic components of kronecker product of bipartite graphs, *Discuss. Math.-Graph Theory* 17 (1997), 301–309.
- [5] P. K. Jha, An counterexample to Tang and Padubidri's claim about bisection of a diagonal mesh, *IEEE Trans. Computers* 52 (2003), 676–677.
- [6] S. Kim, Centers of a tensor composite graph, *Congressus Numerantium* 81 (1991), 193–204.
- [7] K. W. Tang and S. A. Padubidri, Diagonal and toroidal mesh networks, *IEEE Trans. Computers* 43 (1994), 815–826.
- [8] P. Weichsel, The Kronecker product of graphs, *Proc. Amer. Math. Soc.* 13 (1962), 47–52.